

2022 Ph H2 Q1

(a)(i) A)

Calculate the horizontal component of the initial velocity

Given:

- $v = 17.0\text{ m s}^{-1}$
- $\theta = 24.0^\circ$

$v_x = v \cos \theta = 17.0 \times \cos(24.0^\circ) = 17.0 \times 0.9135 = 15.5\text{ m s}^{-1}$

(a)(i) B)

Calculate the vertical component of the initial velocity

$v_y = v \sin \theta = 17.0 \times \sin(24.0^\circ) = 17.0 \times 0.4067 = 6.91\text{ m s}^{-1}$

(a)(ii)

Show that the time taken to reach the crossbar is 0.71 s

Horizontal distance: $d = 11.0\text{ m}$

$t = \frac{d}{v_x} = \frac{11.0}{15.5} = 0.71\text{ s}$

✔ Value confirmed.

(a)(iii)

Calculate the height of the football at the crossbar

Use vertical motion equation:

$$s = v_y t - \frac{1}{2}gt^2$$
$$s = (6.91)(0.71) - \frac{1}{2}(9.8)(0.71)^2$$
$$= 4.91 - 2.47 = 2.44\text{ m}$$

(b)

The ball is kicked at the same angle but with less than 17.0 m/s. What happens?

✔ The ball will pass below the crossbar.

Justification:

- Lower speed = smaller vertical and horizontal components
- So, it won't rise as high, and it will take longer to reach the crossbar
- Result: height at that point will be less than 2.44 m

🧠 Revision Tips

- Use $v_x = v \cos \theta$ and $v_y = v \sin \theta$ to break velocity into components
- Horizontal motion: constant speed → use $s = vt$
- Vertical motion: acceleration due to gravity → use $s = ut + \frac{1}{2}at^2$ or $v^2 = u^2 + 2as$
- For projectile questions, treat horizontal and vertical motion **separately**
- A sketch is often helpful for visualising launch angle, path, and known values

